# • Mini Project Report on Customer Segmentation using K-Means Clustering

## Aim:

To perform customer segmentation using K-Means clustering and visualize groups based on income and spending behavior using R.

## Scenario / Background:

- A retail store collects data about its customers including their annual income and spending habits.

  The management wants to:
  - Identify different types of customers based on purchasing behavior.
  - Group customers into clusters for targeted marketing strategies.
  - Understand high-value and low-value customer segments.
  - Improve business decisions by analyzing customer patterns

## Sample Dataset:

| Customer ID | Income | Spending Score |
|---|---|---|
| 1 | 15000 | 20 |
| 2 | 18000 | 25 |
| 3 | 25000 | 30 |
| 4 | 30000 | 35 |
| 5 | 45000 | 50 |
| 6 | 50000 | 55 |
| 7 | 65000 | 65 |
| 8 | 70000 | 70 |

## R Program (with line-by-line comments):

```
# Create dataset (as given)
 customer <- data.frame(
  CustomerID = 1:8,
  Income = c(15000, 18000, 25000, 30000, 45000, 50000, 65000, 70000),
  Spending = c(20, 25, 30, 35, 50, 55, 65, 70)
)
# Display dataset
print(customer)
# Select required columns
data <- customer[, c("Income", "Spending")]
# Normalize the data
data_scaled <- scale(data)
# Apply K-Means clustering
set.seed(123)
kmeans_result <- kmeans(data_scaled, centers = 3)
# Add cluster labels to dataset
customer$Cluster <- kmeans_result$cluster
# Display result
print(customer)
# Plot clusters
plot(customer$Income, customer$Spending,
     col = customer$Cluster,
     pch = 19,
     main = "Customer Segmentation using K-Means",
     xlab = "Income",
```

```
    ylab = "Spending Score")
# Add cluster centers
centers <- kmeans_result$centers
centers_original <- t(apply(centers, 1, function(x)
  x * attr(data_scaled, "scaled:scale") + attr(data_scaled, "scaled:center")))
points(centers_original,
       col = 1:3,
       pch = 8,
       cex = 2)
# Add legend
legend("topleft",
       legend = c("Cluster 1", "Cluster 2", "Cluster 3"),
       col = 1:3,
       pch = 19)
```

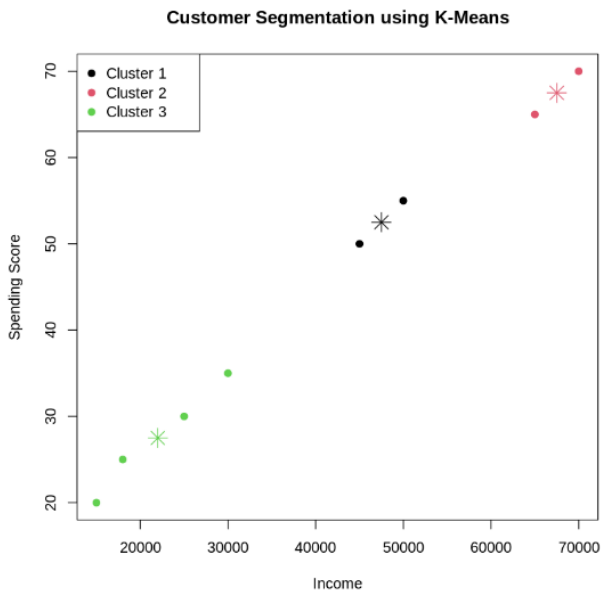
## Output / Observation:

| CustomerID | Income | Spending |
|---|---|---|
| 1 | 15000 | 20 |
| 2 | 18000 | 25 |
| 3 | 25000 | 30 |
| 4 | 30000 | 35 |
| 5 | 45000 | 50 |
| 6 | 50000 | 55 |
| 7 | 65000 | 65 |
| 8 | 70000 | 70 |

| Customer ID | Income | Spending | Cluster |
|---|---|---|---|
| 1 | 15000 | 20 | 3 |

| | | | |
|---|---|---|---|
| 2 | 18000 | 25 | 3 |
| 3 | 25000 | 30 | 3 |
| 4 | 30000 | 35 | 3 |
| 5 | 45000 | 50 | 1 |
| 6 | 50000 | 55 | 1 |
| 7 | 65000 | 65 | 2 |
| 8 | 70000 | 70 | 2 |

## Graph:




## Overall Insight:

Customer segmentation helps businesses target different customer groups effectively and improve decision making.

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CSE, II Year & IV Sem

Topic: **Mini Project Report on Customer Segmentation using K-Means Clustering**

Data Science Using R Programming Lab